

BMAT202L PROBABILITY & STATISTICS

Module 1 Introduction to Statistics

Syllabus

Introduction to Statistics:

Introduction to statistics and data analysis-Measures of central tendency – Measures of variability- [Moments-Skewness-Kurtosis (Concepts only)].

Statistics:

Statistics is the branch of science where we plan, gather and analyze information about a particular collection of individuals or objects under investigation.

Statistics is defined differently by different authors over a period of time.

- Statistics are numerical statement of facts in any department of enquiry placed in relation to each other.

- A.L. Bowley

- Statistics may be defined as the science of collection, presentation analysis and interpretation of numerical data from the logical analysis. It is clear that the definition of statistics by Croxton and Cowden is the most scientific and realistic one. According to this definition there are four stages: Collection of Data, Presentation of data, Analysis of data and Interpretation of data.

- Croxton and Cowden

Basic Steps in a Statistical Study:

For any statistical study, there are some basic steps to be followed once we draw a sample. These are:

- **Step 1:** Gather first-hand information from the sample and this is called the raw data.
- **Step 2:** Tabular representation of the raw data, i.e., represent the raw data in a table.
- **Step 3:** Pictorial representation of the data, i.e., draw diagrams with the organized data in a table.
- **Step 4:** Numerically summarize the data, i.e., describe the entire data set with some key numbers.
- **Step 5:** Analyze the data using mathematical formulae.
- **Step 6:** Draw the final inference or conclusion about the population under study.

Data Analysis:

- The data can be collected in connection with time or geographical location or in connection with time and location.
- Any statistical data can be classified under two categories depending upon the sources utilized.
- Primary data
- Secondary data

Primary Data:

Primary data is the one, which is collected by the investigator himself for the purpose of a specific inquiry or study. Such data is original in character and is generated by survey conducted by individuals or research institution or any organisation

Example:

If a researcher is interested to know the impact of noon meal scheme for the school children, he has to undertake a survey and collect data on the opinion of parents and children by asking relevant questions. Such a data collected for the purpose is called primary data.

Methods for Collecting Primary Data:

The primary data can be collected by the following five methods.

1. Direct personal interviews
2. Indirect Oral interviews
3. Information from correspondents
4. Mailed questionnaire method
5. Schedules sent through enumerators

Secondary Data:

Secondary data are those data which have been already collected and analyzed by some earlier agency for its own use; and later the same data are used by a different agency.

Frequency Distribution:

Frequency distribution is a series when a number of observations with similar or closely related values are put in separate bunches or groups, each group being in order of magnitude in a series. It is simply a table in which the data are grouped into classes and the number of cases which fall in each class are recorded. It shows the frequency of occurrence of different values of a single Phenomenon.

A frequency distribution is constructed for three main reasons:

- 1) To facilitate the analysis of data.
- 2) To estimate frequencies of the unknown population distribution from the distribution of sample data.
- 3) To facilitate the computation of various statistical measures.

Raw Data or Ungrouped Data:

The statistical data collected are generally raw data or ungrouped data.

Example:

Let us consider the daily wages (in Rs.) of 30 laborers in a factory.

800, 700, 550, 500, 600, 650, 400, 300, 800, 900, 750, 450, 350, 650,
700, 800, 820, 550, 650, 800, 600, 550, 380, 650, 750, 850, 900, 650,
450, 750.

Given a raw data set, we can rearrange it in two different ways.

❖ **Frequency distribution or Discrete frequency distribution:**

Using the frequency of the variable we can arrange it. This representation of the data is known as frequency distribution.

❖ **Grouped frequency distribution / Continuous frequency distribution:**

Again we can arrange it for the class intervals. For this situation, it is called as Grouped frequency distribution of the variable.

Examples:

x	f
15	2
17	3
18	5
20	4
22	7
25	9
30	3

(i)

**Frequency distribution
Or
Discrete frequency distribution**

x	f
1 - 9	3
10 - 19	5
20 - 29	10
30 - 39	4
40 - 49	7
50 - 59	6
60 - 69	3

(ii)

Grouped Frequency distribution

x	f
5 - 10	2
10 - 15	3
15 - 20	5
20 - 25	4
25 - 30	7
30 - 35	9
35 - 40	3

(iii)

Continuous Frequency distribution

Special case in grouped frequency distribution:

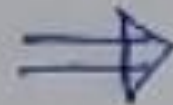
If “ d ” is the gap between the upper limit of any class and the lower limit of the succeeding class, the class boundaries for any class are then given by:

$$\text{Upper class boundary} = \text{Upper class limit} + \frac{d}{2}$$

$$\text{Lower class boundary} = \text{Lower class limit} - \frac{d}{2}$$

Example:-

x	f
15-19	9
20-24	11
25-29	10
30-34	30
35-39	40



x	f
14.5-19.5	9
19.5-24.5	11
24.5-29.5	10
29.5-34.5	30
34.5-39.5	40

Summarizing a raw data set or an organized data set

There are two basic properties of a quantitative data set that are commonly studied. These are central tendency and variability (or dispersion).

Central Tendency: Quite often it is found that the entries in data set cluster around a central (or middle) value. This behavior of the data set is called the central tendency. The main Challenge is to locate a central value around which the clustering takes place.

Three standard methods to measure the location of central tendency are:

- * Mean
- * Median
- * Mode

Variability or Dispersion: Variability or dispersion of data set means the amount of discrepancies among the data entries. There are several ways to measure dispersion or variability in a data set and these are:

- * Range
- * Quartile deviation
- * Variance
- * Standard deviation

Measures of Central Tendency

List of Measures of Central Tendency:

1. Arithmetic Mean or Average
2. Median
3. Mode
4. Geometric Mean
5. Harmonic Mean

1. Arithmetic Mean or Average

❖ Mean of “ n ” observations $(x_1, x_2, \dots, x_n)$ is given by

$$\bar{x} = \frac{1}{n} \sum x_i .$$

❖ In case of the discrete frequency distribution:

If f_i 's are the frequencies of the variable x_i 's then mean

$$\bar{x} = \frac{1}{n} \sum f_i x_i , \quad \text{where } n = \sum f_i .$$

❖ In case of the continuous frequency distribution:

If f_i 's are the frequencies of the class intervals then mean

$$\bar{x} = \frac{\sum (\text{midvalue of each class}) \times f_i}{\sum f_i} .$$

Example:

Find the arithmetic mean of the following frequency distribution:

<i>x</i> :	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>
<i>f</i> :	<i>5</i>	<i>9</i>	<i>12</i>	<i>17</i>	<i>14</i>	<i>10</i>	<i>6</i>

Solution:

<i>x</i>	<i>f</i>	<i>fx</i>
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
	<u>73</u>	<u>299</u>

$$\bar{x} = \frac{1}{N} \Sigma f_x = \frac{299}{73} = 4.09$$

Example:

Calculate the arithmetic mean of the marks from the following table :

Marks	: 0-10	10-20	20-30	30-40	40-50	50-60
No. of students	: 12	18	27	20	17	6

Solution:

Marks	No. of students (f)	Mid - point (x)	fx
0-10	12	5	60
10-20	18	15	270
20-30	27	25	675
30-40	20	35	700
40-50	17	45	765
50-60	6	55	330
Total	100		2,800

$$\text{Arithmetic mean or } \bar{x} = \frac{1}{N} \Sigma fx = \frac{1}{100} \times 2,800 = 28$$

Calculation of mean by using deviation concept:

Sometime the values of the variable (x) or frequency (f) or both are large. Then, the calculation of mean by previous formulas is quite time-consuming. Hence, to avoid such situations we are calculating mean by taking the deviations of the given values from any arbitrary point “ A ” as explained below:

Discrete frequency distribution: $\bar{x} = A + \frac{1}{N} \sum f_i d_i$,

where “ A ” is an arbitrary point, $d_i = x_i - A$ and $N = \sum f_i$.

Continuous frequency distribution: $\bar{x} = A + \frac{h}{N} \sum f_i d_i$,

where “ A ” is an arbitrary point, “ h ” is the magnitude of class interval and $N = \sum f_i$. Here, $d_i = \frac{x_i - A}{h}$, where x_i ’s are the mid value of each class.

Example:

Calculate the mean for the following frequency distribution.

Class-interval :	0–8	8–16	16–24	24–32	32–40	40–48
Frequency :	8	7	16	24	15	7

Solution:

Class-interval	mid-value (x)	Frequency (f)	$d = (x - A) / h$	fd
0–8	4	8	–3	–24
8–16	12	7	–2	–14
16–24	20	16	–1	–16
24–32	28	24	0	0
32–40	36	15	1	15
40–48	44	7	2	14
		77		–25

Here we take $A = 28$ and $h = 8$.

$$\therefore \bar{x} = A + \frac{h \sum fd}{N} = 28 + \frac{8 \times (-25)}{77} = 28 - \frac{200}{77} = 25.404$$

Median

- Median of distribution is the value of the variable which divides it into two equal parts
- It is the value which exceeds and is exceeded by the same number of observations
- Median is the value such that the number of observations above it is equal to the number of observations below it
- The median is thus a positional average

- In case of ungrouped data, if the number of observations is odd then median is the middle value after the values have been arranged in ascending or descending order of magnitude.
- In case of even number of observations, there are two middle terms and median is obtained by taking the arithmetic mean of the middle terms.
- For example, the median of the value 25, 20, 15, 35, 18, i.e., 15, 18, 20, 25, 35 is 20
- The median of 8, 20, 50, 25, 15, 30, i.e., of 8, 15, 20, 25, 30, 50 is $\frac{1}{2}(20 + 25) = 22.5$.

Median for Discrete Frequency Distribution

In case of discrete frequency distribution median is obtained by considering the cumulative frequencies. The steps for calculating median are given below:

- (i) Find $N/2$, where $N = \sum_i f_i$.
- (ii) See the (less than) cumulative frequency (c.f.) just greater than $N/2$.
- (iii) The corresponding value of x is median.

Example:

Obtain the median for the following frequency distribution:

$x :$	1	2	3	4	5	6	7	8	9
$f :$	8	10	11	16	20	25	15	9	6

Solution:

x	f	$c.f.$
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	90
7	15	105
8	9	114
9	6	120
		120

Hence $N = 120 \Rightarrow N/2 = 60$

Cumulative frequency (c.f.) just greater than $N/2$, is 65 and the value of x corresponding to 65 is 5. Therefore, median is 5.

Median for Continuous Frequency Distribution

In the case of continuous frequency distribution, the class corresponding to the *c.f.* just greater than $N/2$ is called the *median class* and the value of median is obtained by the following formula :

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

where l is the lower limit of the median class,

f is the frequency of the median class,

h is the magnitude of the median class,

' c ' is the *c.f.* of the class *preceding* the median class,

and $N = \Sigma f$.

Example:

Find the median wage of the following distribution :

Wages (in Rs.) :	20—30	30—40	40—50	50—60	60—70
No. of labourers :	3	5	20	10	5

Solution:

Wages (in Rs.)	No. of labourers	c.f.
20—30	3	3
30—40	5	8
40—50	20	28
50—60	10	38
60—70	5	43

Here $N/2 = 43/2 = 21.5$. Cumulative frequency just greater than 21.5 is 28 and the corresponding class is 40—50. Thus median class is 40—50.

$$\text{Median} = 40 + \frac{10}{20} (21.5 - 8) = 40 + 6.75 = 46.75$$

Thus median wage is Rs. 46.75.

Example

An incomplete frequency distribution is given as follows

Variable	Frequency	Variable	Frequency
10–20	12	50–60	?
20–30	30	60–70	25
30–40	?	70–80	18
40–50	65	Total	229

Given that the median value is 46, determine the missing frequencies using the median formula.

Solution: Let the frequency of the class 30–40 be f_1 and that of 50–60 f_2 . Then,

$$f_1 + f_2 = 229 - (12 + 30 + 65 + 25 + 18) = 79.$$

Since median is given to be 46, the class 40–50 is the median class. Hence using median formula

we get

$$46 = 40 + \frac{114.5 - (12 + 30 + f_1)}{65} \times 10$$

which gives $f_1 = 34$ and $f_2 = 45$, since frequency never be fractional and $f_1 + f_2 = 79$.

Mode

- Mode is the value which occurs most frequently in a set of observations and around which the other items of the set cluster densely.
- Mode is the value of the variable which is predominant in the series.

Definition (Discrete frequency distribution)

In the case of discrete frequency distribution mode is the value of x corresponding to maximum frequency.

Mode for Discrete Frequency Distribution

In case of discrete frequency distribution
mode is the value of x corresponding to maximum frequency.

Example:

For the following discrete frequency distribution,

x	:	1	2	3	4	5	6	7	8
f	:	4	9	16	25	22	15	7	3

the value of x corresponding to the maximum frequency, viz., 25 is 4. Hence mode is 4.

Mode for Discrete Frequency Distribution

But in any one (or more) of the following cases :

- (i) if the maximum frequency is repeated,
 - (ii) if the maximum frequency occurs in the very beginning or at the end of the distribution, and
 - (iii) if there are irregularities in the distribution,
- the value of mode is determined by the *method of grouping*, which is illustrated below by an example.

Mode for Discrete Frequency Distribution

Method of Grouping

Example:

Find the mode of the following frequency distribution :

Size (x) :	1	2	3	4	5	6	7	8	9	10	11	12
Frequency (f) :	3	8	15	23	35	40	32	28	20	45	14	6

Solution:

Here we see that the distribution is not regular since the frequencies are increasing steadily up to 40 and then decrease but the frequency 45 after 20 does not seem to be consistent with the distribution. Here we cannot say that since maximum frequency is 45, mode is 10. Here we shall locate mode by the method of grouping as explained below :

The frequencies in column (i) are the original frequencies. Column (ii) is obtained by combining the frequencies two by two. If we leave the first frequency and combine the remaining frequencies two by two we get column (iii). Combining the frequencies two by two after leaving the first two frequencies results in a repetition of column (ii). Hence, we proceed to combine the frequencies three by three, thus getting column (iv). The combination of frequencies three by three after leaving the first frequency results in column (v) and after leaving the first two frequencies results in column (vi).

Solution (Continued):

Size (x)	Frequency					
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
1	3	11	23	26	46	73
2	8					
3	15					
4	23	38	58	98	107	100
5	35					
6	40					
7	32	60	48	80	93	79
8	28					
9	20					
10	45	65	59	65		
11	14					
12	6					

Solution (Continued):

The maximum frequency in each column is given in black type. To find mode we form the following table :

ANALYSIS TABLE

<i>Column Number (1)</i>	<i>Maximum Frequency (2)</i>	<i>Value or combination of values of x giving max. frequency in (2) (3)</i>
(i)	45	10
(ii)	75	5, 6
(iii)	72	6, 7
(iv)	98	4, 5, 6,
(v)	107	5, 6, 7
(vi)	100	6, 7, 8

On examining the values in column (3) above, we find that the value 6 is repeated the maximum number of times and hence the value of mode is 6 and not 10 which is an irregular item.

Mode for Continuous Frequency Distribution

In case of continuous frequency distribution, mode is given by the formula :

$$\text{Mode} = l + \frac{h(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

where l is the lower limit, h the magnitude and f_1 the frequency of the modal class, f_0 and f_2 are the frequencies of the classes preceding and succeeding the modal class respectively.

Example:

Find the mode for the following distribution :

<i>Class - interval :</i>	<i>0-10</i>	<i>10-20</i>	<i>20-30</i>	<i>30-40</i>	<i>40-50</i>	<i>50-60</i>	<i>60-70</i>	<i>70-80</i>
<i>Frequency :</i>	<i>5</i>	<i>8</i>	<i>7</i>	<i>12</i>	<i>28</i>	<i>20</i>	<i>10</i>	<i>10</i>

Solution:

Here maximum frequency is 28. Thus the class 40-50 is the modal class.
the value of mode is given by

$$\text{Mode} = 40 + \frac{10(28 - 12)}{(2 \times 28 - 12 - 20)} = 40 + 6.666 = 46.67 \text{ (approx.)}$$

4. Geometric Mean (G)

Geometric mean of a set of n observations is the n th root of their product. Thus the geometric mean G , of n observations $x_i, i = 1, 2, \dots, n$ is

$$G = (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n}$$

The computation is facilitated by the use of logarithms. Taking logarithm of both sides, we get

$$\log G = \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n) = \frac{1}{n} \sum_{i=1}^n \log x_i$$

$$\therefore G = \text{Antilog} \left[\frac{1}{n} \sum_{i=1}^n \log x_i \right]$$

4. Geometric Mean (Continued)

In case of frequency distribution $x_i | f_i$, ($i = 1, 2, \dots, n$) geometric mean, G is given by

$$G = \left[x_1^{f_1} \cdot x_2^{f_2} \cdot \dots \cdot x_n^{f_n} \right]^{\frac{1}{N}}, \text{ where } N = \sum_{i=1}^n f_i$$

Taking logarithms of both sides, we get

$$\begin{aligned} \log G &= \frac{1}{N} (f_1 \log x_1 + f_2 \log x_2 + \dots + f_n \log x_n) \\ &= \frac{1}{N} \sum_{i=1}^n f_i \log x_i \end{aligned}$$

Thus we see that logarithm of G is the arithmetic mean of the logarithms of the given values.

$$G = \text{Antilog} \left(\frac{1}{N} \sum_{i=1}^n f_i \log x_i \right)$$

In the case of grouped or continuous frequency distribution, x is taken to be the value corresponding to the mid-point of the class-intervals.

5. Harmonic Mean (H)

Harmonic mean of a number of observations is the reciprocal of the arithmetic mean of the reciprocals of the given values. Thus, harmonic mean H , of n observations x_i , $i = 1, 2, \dots, n$ is

$$H = \frac{1}{\frac{1}{n} \sum_{i=1}^n (1/x_i)}$$

In case of frequency distribution $x_i | f_i$, ($i = 1, 2, \dots, n$),

$$H = \frac{1}{\frac{1}{N} \sum_{i=1}^n (f_i/x_i)}, \quad \left[N = \sum_{i=1}^n f_i \right]$$

Karl Pearson Relationship:

Sometimes mode is estimated from the mean and the median. For a symmetrical distribution, mean, median and mode coincide. If the distribution is moderately asymmetrical, the mean, median and mode obey the following empirical relationship (due to Karl Pearson) :

The distance between mean and median is about one-third of the distance between the mean and mode

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

Which gives, $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}.$

Relation between Mean, Median, Mode:

1. In symmetrical distribution $Mean = Median = Mode$.
2. In positively skewed distribution $Mode < Median < Mean$.
3. In negatively skewed distribution $Mean < Median < Mode$.

Measures of Variability or Measures of Dispersion

Dispersion - Variations or Scatteredness

List of Measures of Dispersion:

1. Range
2. Quartile Deviation or Semi-interquartile Range
3. Mean Deviation and
4. Standard Deviation

Partitions:

- Quartiles
- Deciles
- Percentiles

1. Range

Range = Maximum Value - Minimum Value

The range is the difference between two extreme observations of the distribution. If A and B are the greatest and smallest observations respectively in a distribution, then its range is $A - B$.

Range is the simplest but a crude measure of dispersion. Since it is based on two extreme observations which themselves are subject to chance fluctuations, it is not at all a reliable measure of dispersion.

Partitions:

These are the values which divided the series into a number of equal parts.

Quartiles: The three points which divided the series in to four equal parts are called quartiles. It is denoted by Q_1, Q_2, Q_3 .

Deciles: The nine points which divided the series in to ten equal parts are called deciles. It is denoted by $D_1, D_2, \dots, D_9$.

Percentiles: The ninety-nine points which divided the series in to hundred equal parts are called percentiles. It is denoted by $P_1, P_2, \dots, P_{99}$.

For discrete frequency distribution:

Quartiles:- $Q_k : \frac{kN}{4}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{4}$, the corresponding variable is the quartile value. Here, $k = 1, 2, 3$.

Deciles:- $D_k : \frac{kN}{10}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{10}$, the corresponding variable is the decile value. Here, $k = 1, 2, 3, 4, 5, 6, 7, 8, 9$.

Percentiles:- $P_k : \frac{kN}{100}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{100}$, the corresponding variable is the percentile value. Here, $k = 1, 2, \dots, 99$.

For continuous frequency distribution:

Quartiles:- Step-1: Find quartile class by:

Compute $\frac{kN}{4}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{4}$.

Step-2: Use the formula

$$Q_k = l + \frac{h}{f} \left(\frac{kN}{4} - c \right) ; k = 1, 2, 3.$$

Deciles:- Step-1: Find decile class by:

Compute $\frac{kN}{10}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{10}$.

Step-2: Use the formula

$$D_k = l + \frac{h}{f} \left(\frac{kN}{10} - c \right) ; k = 1, 2, \dots, 9.$$

Percentiles:- Step-1: Find percentile class by:

Compute $\frac{kN}{100}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{100}$.

Step-2: Use the formula

$$P_k = l + \frac{h}{f} \left(\frac{kN}{100} - c \right) ; k = 1, 2, \dots, 99.$$

Quartile Deviation or Semi-interquartile Range (Q)

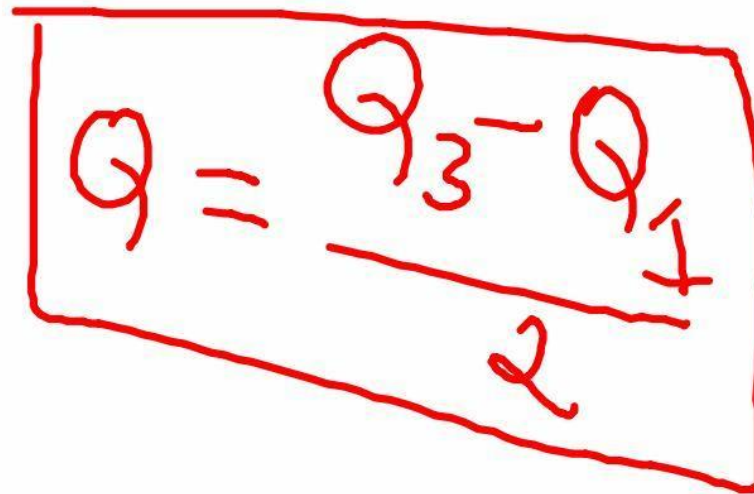
Q is given by

$$Q = \frac{1}{2} (Q_3 - Q_1),$$

where Q_1 and Q_3 are the first and third quartiles of the distribution respectively.

Quartile deviation is definitely a better measure than the range as it makes use of 50% of the data. But since it ignores the other 50% of the data, it cannot be regarded as a reliable measure.

$$\text{Quartile Deviation (Q)} = (1/2)(Q_3 - Q_1).$$


$$Q = \frac{Q_3 - Q_1}{2}$$

Example:

Eight coins were tossed together and the number of heads resulting was noted. The operation was repeated 256 times and the frequencies (f) that were obtained for different values of x , the number of heads, are shown in the following table. Calculate median, quartiles.

$x:$	0	1	2	3	4	5	6	7	8
$f:$	1	9	26	59	72	52	29	7	1

Solution:

$x:$	0	1	2	3	4	5	6	7	8
$f:$	1	9	26	59	72	52	29	7	1
$c.f.:$	1	10	36	95	167	219	248	255	256

Median : Here $N/2 = 256/2 = 128$. Cumulative frequency ($c.f.$) just greater than 128 is 167. Thus, median = 4.

Q_1 : Here $N/4 = 64$. $c.f.$ just greater than 64 is 95. Hence, $Q_1 = 3$.

Q_3 : Here $3N/4 = 192$ and $c.f.$ just greater than 192 is 219. Thus $Q_3 = 5$.

Quartile Deviation (Q) = $(1/2)(Q_3 - Q_1)$.

3. Mean Deviation

If $x_i | f_i, i = 1, 2, \dots, n$ is the frequency distribution, then mean deviation from the average A , (usually mean, median or mode), is given by

$$\text{Mean deviation} = \frac{1}{N} \sum_i f_i |x_i - A|, \quad \sum f_i = N$$

where $|x_i - A|$ represents the modulus or the absolute value of the deviation $(x_i - A)$, when the -ive sign is ignored.

Since mean deviation is based on all the observations, it is a better measure of dispersion than range or quartile deviation. But the step of ignoring the signs of the deviations $(x_i - A)$ creates artificiality and renders it useless for further mathematical treatment.

It may be pointed out here that mean deviation is least when taken from median.

$$MD = \frac{1}{N} \sum_{i=1}^n f_i |x_i - A|, \quad N = \sum_{i=1}^n f_i$$

4. Standard Deviation (σ)

Standard

deviation, usually denoted by the Greek letter small sigma (σ), is the positive square root of the arithmetic mean of the squares of the deviations of the given values from their arithmetic mean. For the frequency distribution $x_i | f_i, i = 1, 2, \dots, n$,

$$\sigma = \sqrt{\frac{1}{N} \sum_i f_i (x_i - \bar{x})^2}$$

$\bar{x} \rightarrow$ Mean

where $\bar{x}$ is the arithmetic mean of the distribution and $\sum_i f_i = N$.

Variance

The square of standard deviation is called the *variance* and is given by

$$\sigma^2 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2$$

Root Mean Square Deviation

Root mean square deviation, denoted by 's' is given by

$$s = \sqrt{\frac{1}{N} \sum_i f_i (x_i - A)^2}$$

Any Number

where A is any arbitrary number. s^2 is called mean square deviation.

Difference Formulae for Calculating Variance

$$\sigma^2 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2$$

$$\sigma_x^2 = \frac{1}{N} \sum_i f_i x_i^2 - \left(\frac{1}{N} \sum_i f_i x_i \right)^2$$

We know that if $d_i = x_i - A$ then $\bar{x} = A + \frac{1}{N} \sum_i f_i d_i$

$$\therefore A - \bar{x} = -\frac{1}{N} \sum_i f_i d_i$$

Hence

$$\sigma_x^2 = \frac{1}{N} \sum_i f_i d_i^2 + \left(-\frac{1}{N} \sum_i f_i d_i \right)^2 + 2 \left(-\frac{1}{N} \sum_i f_i d_i \right) \left(\frac{1}{N} \sum_i f_i d_i \right)$$

$$= \frac{1}{N} \sum_i f_i d_i^2 - \left(\frac{1}{N} \sum_i f_i d_i \right)^2$$

$\Rightarrow$

$$\sigma_x^2 = \sigma_d^2$$

Hence variance and consequently standard deviation is independent of change of origin.

Co-efficient of Dispersion

Whenever we want to compare the variability of the two series which differ widely in their averages or which are measured in different units, we do not merely calculate the measures of dispersion but we calculate the co-efficients of dispersion which are pure numbers independent of the units of measurement. The co-efficients of dispersion (C.D.) based on different measures of dispersion are as follows :

1. C.D. based upon range = $\frac{A - B}{A + B}$, where A and B are the greatest and

the smallest items in the series.

2. Based upon quartile deviation :

$$\text{C.D.} = \frac{(Q_3 - Q_1)/2}{(Q_3 + Q_1)/2} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

3. Based upon mean deviation :

$$\text{C.D.} = \frac{\text{Mean deviation}}{\text{Average from which it is calculated}}$$

4. Based upon standard deviation :

$$\text{C.D.} = \frac{\text{S.D.}}{\text{Mean}} = \frac{\sigma}{\bar{x}}$$

Co-efficient of Variation

100 times the co-efficient of dispersion based upon standard deviation is called co-efficient of variation (C.V.),

$$C.V. = 100 \times \frac{\sigma}{\bar{x}}$$

According to Professor Karl Pearson who suggested this measure, C.V. is *the percentage variation in the mean, standard deviation being considered as the total variation in the mean.*

For comparing the variability of two series, we calculate the co-efficient of variations for each series. The series having greater C.V. is said to be more variable than the other and the series having lesser C.V. is said to be more consistent (or homogenous) than the other.

(2) Relations between measures of dispersion

- (i) *Quartile deviation = 2/3 (standard deviation)*
- (ii) *Mean deviation = 4/5 (standard deviation)*

Example:

Calculate the mean and standard deviation for the following table giving the age distribution of 542 members.

Age in years :	20—30	30—40	40—50	50—60	60—70	70—80	80—90
No. of members :	3	61	132	153	140	51	2

Solution: Also find CD & CV.

$$\text{Here we take } d = \frac{x - A}{h} = \frac{x - 55}{10}$$

Age group	Mid-value (x)	Frequency (f)	$d = \frac{x - 55}{10}$	fd	fd ²
20 — 30	25	3	-3	-9	27
30 — 40	35	61	-2	-122	244
40 — 50	45	132	-1	-132	132
50 — 60	55	153	0	0	0
60 — 70	65	140	1	140	140
70 — 80	75	51	2	102	204
80 — 90	85	2	3	6	18
		$N = \Sigma f = 542$		$\Sigma fd = -15$	$\Sigma fd^2 = 765$

$$\bar{x} = A + h \frac{\Sigma fd}{N} = 55 + \frac{10 \times (-15)}{542} = 55 - 0.28 = 54.72 \text{ years.}$$

$$\sigma^2 = h^2 \left[\frac{1}{N} \Sigma fd^2 - \left(\frac{1}{N} \Sigma fd \right)^2 \right] = 100 \left[\frac{765}{542} - (0.28)^2 \right] \quad \text{CD} = \frac{\sigma}{\bar{x}}$$

$$= 100 \times 1.333 = 133.3$$

$$\sigma \text{ (standard deviation)} = 11.55 \text{ years}$$

$$\text{CV} = 100 \times \frac{\sigma}{\bar{x}}$$

Moments

The r th moment of a variable x about any point $x = A$, usually denoted by μ_r' is given by

$$\begin{aligned}\mu_r' &= \frac{1}{N} \sum_i f_i (x_i - A)^r, \quad \sum_i f_i = N \\ &= \frac{1}{N} \sum_i f_i d_i^r,\end{aligned}$$

where $d_i = x_i - A$.

The r th moment of a variable about the mean $\bar{x}$, usually denoted by μ_r , is given by

$$\mu_r = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^r = \frac{1}{N} \sum_i f_i z_i^r$$

where

$$z_i = x_i - \bar{x}.$$

$$\mu_r' = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r$$

$$A = \bar{x}$$

$$\mu_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r$$

Particular Cases

In particular

$$\mu_0 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^0 = \frac{1}{N} \sum_i f_i = 1$$

and $\mu_1 = \frac{1}{N} \sum_i f_i (x_i - \bar{x}) = 0$, being the algebraic sum of deviations from the mean. Also

$$\mu_2 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2 = \sigma^2$$

These results, viz., $\mu_0 = 1$, $\mu_1 = 0$, and $\mu_2 = \sigma^2$, are of fundamental importance and should be committed to memory.

Pearson's Co-efficients

Karl Pearson defined the following

four coefficients, based upon the first four moments about mean :

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \quad \gamma_1 = + \sqrt{\beta_1} \quad \text{and} \quad \beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_2 = \beta_2 - 3$$


Skewness

Literally, skewness means '*lack of symmetry*'. We study skewness to have an idea about the shape of the curve which we can draw with the help of the given data. A distribution is said to be skewed if

- (i) Mean, median and mode fall at different points,
i.e., $\text{Mean} \neq \text{Median} \neq \text{Mode}$,
- (ii) Quartiles are not equidistant from median, and
- (iii) The curve drawn with the help of the given data is not symmetrical but stretched more to one side than to the other.

Measures of Skewness

Measures of Skewness. Various measures of skewness are

$$(1) S_k = M - M_d \quad (2) S_k = M' - M_0.$$

where M is the mean, M_d , the median and M_0 , the mode of the distribution.

$$(3) S_k = (Q_3 - M_d) - (M_d - Q_1).$$

These are the absolute measures of skewness. As in dispersion, for comparing two series we do not calculate these absolute measures but we calculate the relative measures called the *co-efficients of skewness* which are pure numbers independent of units of measurement.

Co-efficients of Skewness

1. *Prof. Karl Pearson's Coefficient of Skewness.*

$$S_k = \frac{(M - M_0)}{\sigma}$$

II. *Prof. Bowley's Coefficient of Skewness.* Based on quartiles,

$$S_K = \frac{(Q_3 - M_d) - (M_d - Q_1)}{(Q_3 - M_d) + (M_d - Q_1)} = \frac{Q_3 + Q_1 - 2M_d}{Q_3 - Q_1}$$

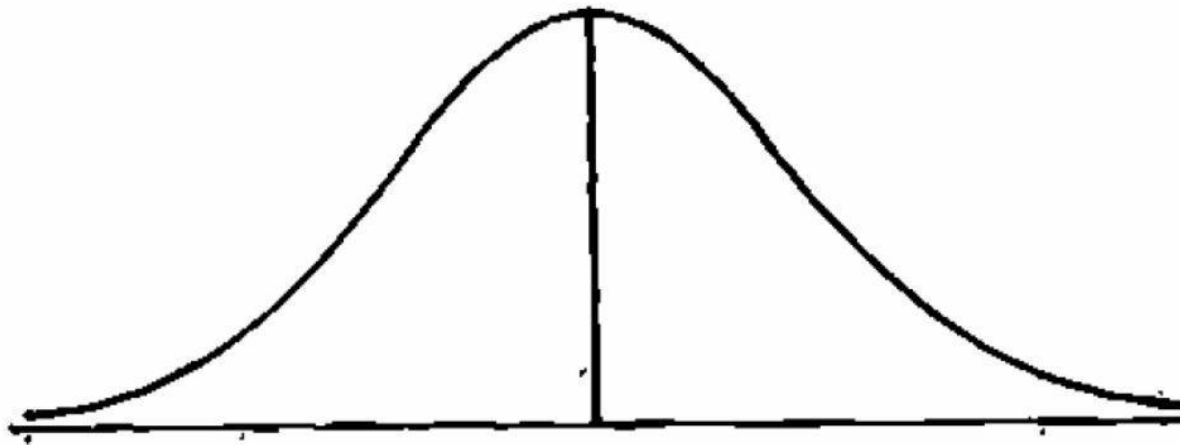
III. Based upon moments, co-efficient of skewness is

$$S_k = \frac{\sqrt{\beta_1} (\beta_2 + 3)}{2 (5\beta_2 - 6\beta_1 - 9)}$$

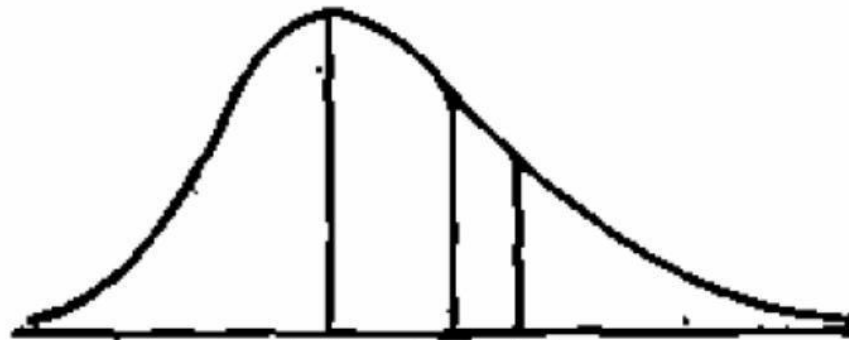
where symbols have their usual meaning. Thus $S_k = 0$ if either $\beta_1 = 0$ or $\beta_2 = -3$. But since $\beta_2 = \mu_4/\mu_2^2$, cannot be negative, $S_k = 0$ if and only if $\beta_1 = 0$. Thus for a symmetrical distribution $\beta_1 = 0$. In this respect β_1 is taken to be a measure of skewness.

The skewness is positive if the larger tail of the distribution lies towards the higher values of the variate (the right), i.e., if the curve drawn with the help of the given data is stretched more to the right than to the left and is negative

Graphical Representation of Skewness

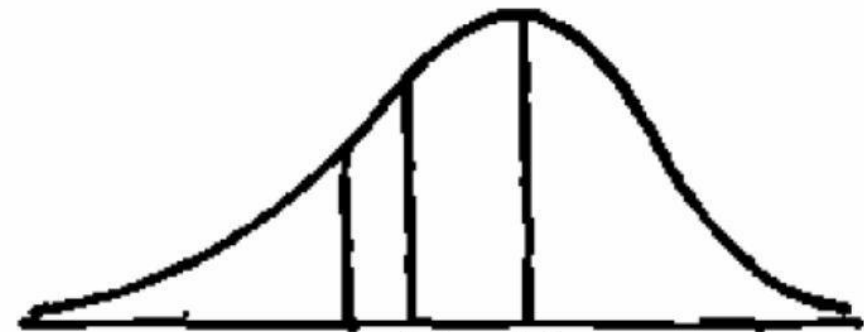


$\bar{x}$ (Mean) = M_0 = M_d
(Symmetrical Distribution)



M_0 M_d Mean

(Positively Skewed Distribution)



Mean M_d M_0

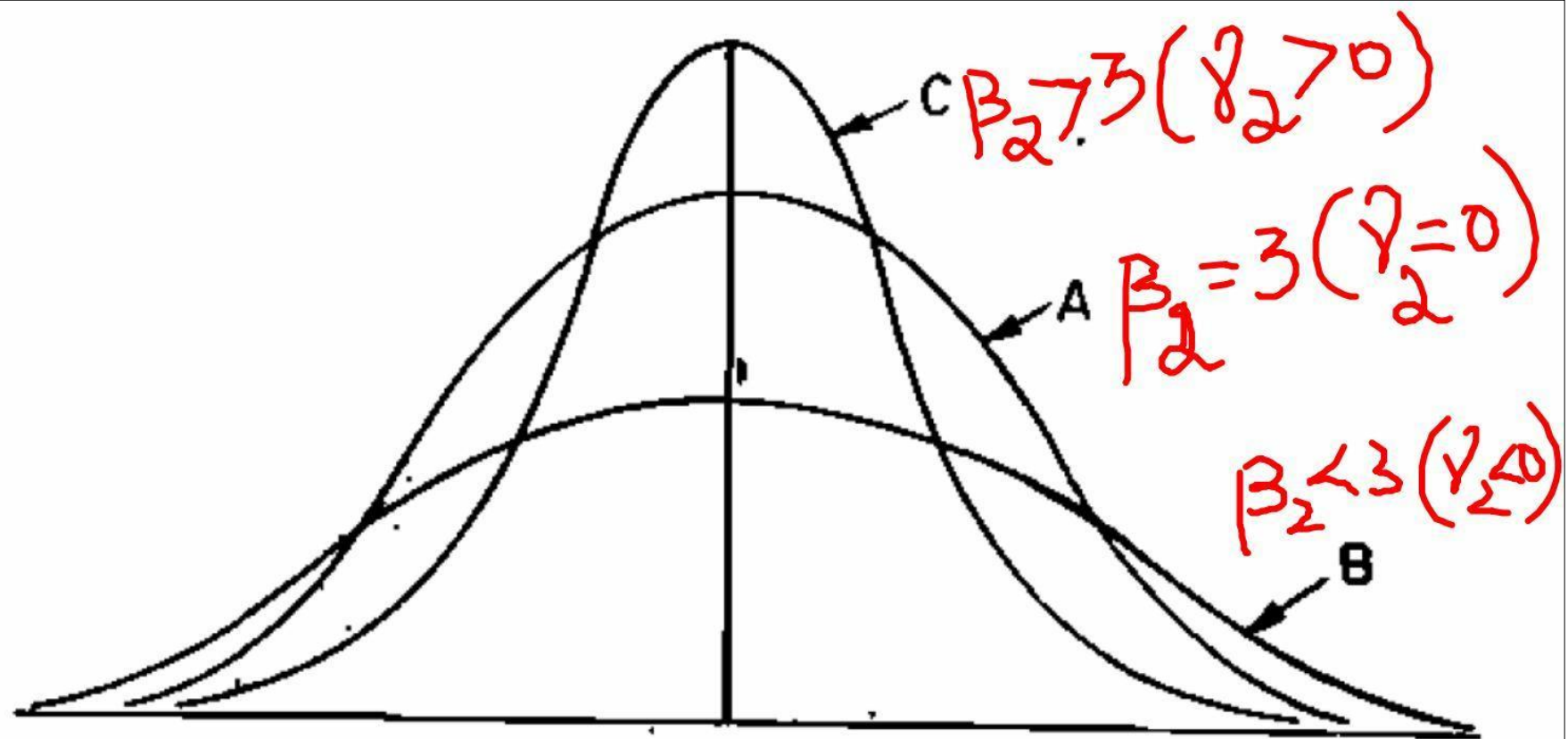
(Negatively Skewed Distribution)

Kurtosis

If we know the measures of central tendency, dispersion and skewness, we still cannot form a complete idea about the distribution as will be clear from the following figure in which all the three curves A, B and C are symmetrical about the mean 'm' and have the same range.

In addition to these measures we should know one more measure which Prof. Karl Pearson calls as the 'Convexity of curve' or *Kurtosis*. Kurtosis enables us to have an idea about the flatness or peakedness of the curve. It is measured by the co-efficient β_2 or its derivation γ_2 given by

$$\underline{\beta_2 = \mu_4/\mu_2^2, \quad \gamma_2 = \beta_2 - 3}$$



Curve of the type 'A' which is neither flat nor peaked is called the *normal curve* or *mesokurtic curve* and for such a curve $\beta_2 = 3$, i.e., $\gamma_2 = 0$. Curve of the type 'B' which is flatter than the normal curve is known as *platykurtic* and for such a curve $\beta_2 < 3$, i.e., $\gamma_2 < 0$. Curve of the type 'C' which is more peaked than the normal curve is called *leptokurtic* and for such a curve, $\beta_2 > 3$, i.e., $\gamma_2 > 0$.



Problems based on the Moments, Skewness and Kurtosis concepts:

- Q1) Find the first four moments about $x = 10$ for the series 4, 7, 10, 13, 16, 19, 22.
- Q2) Calculate the first four moments about the mean for the series 4, 7, 10, 13, 16, 19, 22.
- Q3) The first four moments of a distribution about $x = 4$ are 1, 4, 10 and 45. Comment upon the nature of the distribution.
- Q4) In a certain distribution, the first four moments about $x=5$ are 2, 20, 40 and 50. Calculate and state whether the distribution is leptokurtic or platykurtic.
- Q5) The first four central moments of a distribution are 0, 2.5, 0.7 and 18.75. Test the skewness and kurtosis of the distribution.

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